BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . MEDMASTERS COLLABORATIVE

Module 12: Cardiovascular System

Week 15 teaching guide . 3 topics

This is the instructor-facing companion to the student pre-work hub. For every topic, you have the science to teach, the recommended approach to learning that material, and why it matters to your students' future patients. Lecture order, common misconceptions, and pre-video drawing prompts are baked in so each video lesson can hit the same pedagogical beats.

Topics in this module

1. Heart Anatomy and the Cardiac Cycle

Chambers, valves, great vessels, and the pressure-volume story of one heartbeat.

2. Cardiac Conduction System and ECG

How the heart paces itself, and the wave pattern that records it.

3. Blood Vessels and Hemodynamics

Vessel types, blood pressure determinants, and how flow is regulated.

Module 12 of 17 . Dr. Sharilyn Rennie

TOPIC 1 OF 3 . WEEK 15 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Heart Anatomy and the Cardiac Cycle

Chambers, valves, great vessels, and the pressure-volume story of one heartbeat.

The Science

The heart has four chambers and four valves. Right side carries deoxygenated blood: vena cavae into right atrium (RA) through tricuspid valve into right ventricle (RV), through pulmonary valve into pulmonary trunk and lungs. Left side carries oxygenated blood: pulmonary veins into left atrium (LA), through mitral (bicuspid) valve into left ventricle (LV), through aortic valve into aorta. The two sides are functionally in series.

The cardiac cycle has four phases (one beat = one cycle): ventricular filling (AV valves open, blood drops in passively, then atrial systole tops it off), isovolumetric contraction (all valves closed, pressure rises in the ventricles), ventricular ejection (semilunar valves open, blood is ejected), isovolumetric relaxation (all valves closed, pressure falls). The cycle then restarts.

Heart sounds: S1 ('lub') is the closure of the AV valves at the start of systole. S2 ('dub') is the closure of the semilunar valves at the end of systole. S3 (early diastolic gallop) can be normal in young athletes or pathologic in heart failure. S4 (late diastolic) is often pathologic, reflecting a stiff ventricle. Murmurs are audible turbulence; the timing in the cycle and where they are loudest tell you which valve is malfunctioning.

Frank-Starling mechanism: within the working range, the more the ventricle is stretched by incoming blood (preload), the more forcefully it contracts on the next beat. The heart pumps out what it receives. This is the basis of moment-to-moment matching of cardiac output to venous return.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the heart in cross-section, label all four chambers and four valves, and trace the path of a red blood cell from vena cava to aorta.

Common misconceptions to address

- Students think the heart is one pump. It is two pumps in series.
- S1 and S2 are sometimes mixed. Cue: S1 (lub) is louder and longer (AV valves shutting against ventricular pressure); S2 (dub) is sharper.
- Frank-Starling is sometimes thought of as a feedback loop. It is a structural property of muscle, not a regulatory loop.

Order of operations (what to teach first)

Anatomy and blood flow, then valves and the cardiac cycle phases, then heart sounds, then Frank-Starling as the integrating concept.

Self-test prompts (for students before the recall deck)

- Trace blood from the inferior vena cava to the aorta, naming every chamber and valve.
- Which valves close at S1? Which at S2?
- What is happening during isovolumetric contraction?

Why This Matters to Your Patients

Mitral regurgitation: blood leaks back into the LA during systole; forward cardiac output drops; chronic volume overload dilates the LV. Aortic stenosis: narrowed valve raises the pressure the LV must generate; chronic pressure overload causes concentric LV hypertrophy. Heart failure with reduced ejection fraction (HFrEF): the LV ejects less than 40% per beat; backward congestion produces pulmonary edema; forward underperfusion produces fatigue and renal dysfunction. Cardiac tamponade: fluid in the pericardial sac compresses the heart, dropping preload and cardiac output; this is a true emergency. Nurses monitor for the early signs (rising JVP, dropping BP, muffled heart sounds: Beck's triad). Every patient with a murmur deserves a careful reading of timing and location.

TOPIC 2 OF 3 . WEEK 15 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Cardiac Conduction System and ECG

How the heart paces itself, and the wave pattern that records it.

The Science

The heart paces itself. The sinoatrial (SA) node in the right atrial wall is the normal pacemaker, firing at about 60-100 per minute at rest. The signal spreads through atrial myocardium to the atrioventricular (AV) node, which deliberately delays the impulse (about 100 ms) so the atria finish contracting before the ventricles start. Then the signal travels through the Bundle of His, down the right and left bundle branches, and into the Purkinje fibers, which rapidly distribute it to the ventricular myocardium.

The ECG records this electrical activity. P wave = atrial depolarization. QRS complex = ventricular depolarization (atrial repolarization is hidden inside it). T wave = ventricular repolarization. The PR interval includes AV nodal delay; the QT interval spans ventricular depolarization through repolarization. Normal intervals are well defined, and deviations point to specific problems.

Arrhythmias: AV block (first-degree = prolonged PR; second-degree = some dropped beats; third-degree = atria and ventricles dissociated). Atrial fibrillation (chaotic atrial activity, irregularly irregular ventricular response). Ventricular tachycardia and fibrillation (loss of organized ventricular contraction; rapidly fatal without defibrillation).

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the conduction pathway: SA, AV, Bundle of His, bundle branches, Purkinje. Then they sketch the corresponding ECG: P, QRS, T.

Common misconceptions to address

- Students think the AV node speeds the signal up. It deliberately delays it.
- The T wave is sometimes thought to represent atrial repolarization. No, that is hidden inside the QRS; the T wave is ventricular repolarization.
- ECG and EKG are the same thing (German 'kardiogramm' vs English 'cardiogram'); students sometimes think they are different tests.

Order of operations (what to teach first)

Conduction anatomy, then how each part of the pathway shows up on the ECG, then common arrhythmias as failure modes.

Self-test prompts (for students before the recall deck)

- What sets the normal heart rate?
- What does each wave on the ECG represent?
- Why is there an AV nodal delay?

Why This Matters to Your Patients

Atrial fibrillation is the most common sustained arrhythmia in adults; it raises stroke risk because stagnant blood in the fibrillating atrium can form clots that embolize. Anticoagulation (warfarin, DOACs) is standard. Long QT syndrome (congenital or drug-induced) predisposes to torsades de pointes, a polymorphic ventricular tachycardia. Heart blocks may require pacemakers. Ventricular fibrillation is the cardiac arrest rhythm that responds to defibrillation; this is why AEDs are everywhere. Every nurse monitoring a patient on telemetry is reading the ECG for these patterns in real time.

TOPIC 3 OF 3 . WEEK 15 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Blood Vessels and Hemodynamics

Vessel types, blood pressure determinants, and how flow is regulated.

The Science

Blood vessels are a structural and functional gradient. Arteries are thick-walled and elastic, carrying blood under high pressure. Arterioles are smaller and heavily muscled; they are the primary site of vascular resistance. Capillaries are single-endothelial-layer tubes where exchange happens. Venules and veins are thin-walled with valves; they hold most of the blood volume (capacitance vessels).

Blood pressure equation: BP = cardiac output (CO) x total peripheral resistance (TPR). Cardiac output equation: CO = stroke volume (SV) x heart rate (HR). Stroke volume is set by preload, contractility, and afterload. Total peripheral resistance is dominated by arteriolar diameter. Mean arterial pressure (MAP) is the perfusion pressure that matters to tissues; you want it roughly 65 or above.

Regulation: short-term, the baroreflex (carotid sinus and aortic arch baroreceptors to medullary cardiovascular centers) adjusts HR and vascular tone within seconds. Long-term, the kidneys via RAAS (renin-angiotensin-aldosterone) and ANP regulate blood volume. Both axes are clinical workhorses.

Teaching This Topic

Before the video (drawing prompt)

Have students predict what happens to BP if (a) arterioles vasoconstrict, (b) HR doubles, (c) plasma volume drops. Then they have to commit before checking.

Common misconceptions to address

- Students think 'high BP' just means HR is fast. It can also mean high resistance, high volume, high contractility.
- Veins are sometimes called 'low-pressure arteries.' They have different structure: thin walls, valves, more compliant.
- Capillaries are thought of as small arteries. They are unique: single layer, no muscle, designed for exchange.

Order of operations (what to teach first)

Vessel types, then the BP equation and components, then short-term and long-term regulation.

Self-test prompts (for students before the recall deck)

- Where in the circulation is the primary site of resistance?
- Write the equations for cardiac output and blood pressure.
- Why do veins have valves?

Why This Matters to Your Patients

Hypertension is the most common chronic disease in adults. Treatments target each node of the BP equation: diuretics drop volume (and therefore CO), beta blockers drop HR and contractility, calcium channel blockers drop contractility and vasodilate, ACE inhibitors drop RAAS-driven vasoconstriction and volume retention. Hypovolemic shock from blood loss produces tachycardia (baroreflex compensation) and vasoconstriction; nurses watch for these as the warning signs before BP itself collapses. Orthostatic hypotension (BP drops when standing) reflects autonomic or volume problems and is a fall risk in the elderly.